

L-1: Introduction to Nowcasting

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Mr. Paul Cashin
Director
IMF – STI

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Outline

1. Why nowcasting matters, especially in times of crisis?

2. Usefulness of nowcasting for policy making.

3. How is nowcasting used in practice?

Country examples:

Australia, Atlanta Fed, BoE

Motivation

Policy-makers need to assess the current state of the economy in real-time, when only incomplete information is available (e.g., Lewis *et. al.* (2020), Woloszko (2020)).

- ❑ Important policy decisions need to be made when major shocks occur, e.g., the 2008 GFC, COVID-19, climate-related shocks, etc.

→ *Data deficiencies/publication lags present major challenges for making timely policy decisions e.g., unbalanced data sets*

→ *A dilemma faced by forecasters is that the relevant data are not sampled at the same frequency. Also, standard statistical procedures/forecasting tools assume a single frequency.*

Motivation

Main releases	Publishing lag	Frequency
HICP	1 month	monthly
PPI	2 months	monthly
Industrial production	2 months	monthly
Industrial new orders	2 months	monthly
Turnover index	2 months	monthly
Hours worked	2 months	monthly
Car registrations	2 months	monthly
Retail trade	2 months	monthly
Construction output	2 months	monthly
Business survey	current month	monthly
Business climate indicator	current month	monthly
Consumer survey	current month	monthly
Money supply	1 month	monthly
Exchange rates (average)	1 month	monthly
Interest rates (average)	1 month	monthly
Stock exhange indexes (average)	1 month	monthly
Unemployment	2 months	monthly
GDP: disaggregation of sectorial value added	1 quarter	quarterly
GDP: disaggregation from expenditure side	1 quarter	quarterly

Source: Foroni and Marcellino (2014)

Motivation

In real-time, unbalanced datasets arise mainly because of two reasons:

- The different sampling frequencies with which the indicators are available

  Example: a range of leading and coincident indicators is available at a monthly or even higher frequency.

- The "ragged-edge" problem, i.e., publication delays of the relevant indicators, resulting in missing values at the end of the sample for some of the series.

  Example: key indicators of macroeconomic activity (e.g. GDP) are released quarterly and with considerable publication lag

Forecasting at Central Banks

- Forecasting is a crucial foundation for forward-looking monetary policy decisions.
- Forecasting serves many purposes:
  - To present and analyze the macroeconomic outlook
  - To communicate the outlook for the economy and monetary policy to the public
  - To evaluate different monetary policy strategies (e.g., lowering interest rates compared to quantitative easing).
- Analysts usually use more than one method to forecast a series. They also compare/evaluate the separate forecasts for accuracy.

Key features of international central bank forecasts

	Bank of England	Bank of Canada	European Central Bank	Norgesbank	Reserve Bank of New Zealand	Reserve Bank of Australia	Riksbank	Swiss National Bank	Bank of Japan	Federal Open Market Committee
Forecast ownership										
Ownership of forecast	Policymakers	Policymakers	Staff	Policymakers	Policymakers	Policymakers	Policymakers	Policymakers	Policymakers	Policymakers
Range of policymakers' views quantified?	No	No	No	No	No	No	No	No	Yes	Yes
Forecast presentation										
Point forecasts	Yes[1]	Yes	No	Yes	Yes	No	Yes	Yes	No	No
Interval	No	No	Yes	No	No	Yes	No	No	Yes	Yes
Fan charts	Yes	Yes	No	Yes	No	No	Yes	No	Yes[2]	No
Alternative scenarios quantified	No	No	No	Yes[3]	No	No	Yes[4]	No	No	No
Forecast variables										
GDP growth	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Components of GDP	No	Yes	Yes	Yes	Yes	No	Yes	No	No	No
Unemployment	No	No	No	Yes	Yes	No	Yes	No	No	Yes
output gap/potential output	No	Yes	No	Yes	Yes	No	Yes	No	No	No
Wages/labor costs	No	No	No	Yes	Yes	No[5]	Yes	No	No	No
CPI inflation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Narrower measure of inflation[6]	No	Yes	No	Yes	No	Yes	Yes	No	Yes	Yes
Conditioning assumptions for policy										
Unchanged policy rate	Yes	No	No	No	No	Yes[7]	No	Yes[8]	No	No
Market-implied path	Yes	No	Yes	No	No	No	No	No	No	No
Endogenous policy[9] assumed, not published	No	Yes	No	No	No	No	No	No	Yes	No
Endogenous policy[9] assumed, published	No	No	No	Yes	Yes	No	Yes	No	No	Yes
Forecast frequency										
Forecasts per year	4	4	4	3	4	4	6	4	4	4

What is Nowcasting?

"Nowcasts" are forecasts of the here and now ("now" + "forecast" = "nowcast").

→ They allow for real-time forecasting of a lower frequency variable based on other series released at a similar or higher frequency.

Example: Nowcast the current quarter GDP to be released using a combination of daily, weekly, monthly and quarterly data.

- The nowcast could be updated daily – the highest frequency of explanatory data – as new data releases for the series used to explain GDP become available.
- Trade-off between data frequency and forecast stability: daily indicators can be informative but also noisy, and may produce nowcasts of GDP growth that change too much over time.

Why nowcasting matters, especially in times of crisis?

Nowcasts can be updated in real-time, reducing the risks of stale or inaccurate near-term forecasts.

Nowcasts can accommodate changes in the relevant information set. For example, more indicators are typically used during crisis periods than normal times.

• "soft data" (e.g., consumer or business surveys) is most useful early in the nowcasting cycle, and "hard data" (e.g., retail spending, housing or labor market data) is of more use later.

- soft data often "moves" the nowcast while hard data tends to make it "more precise" (i.e., by reducing the width of the confidence interval of the forecast)

Important Caveat

- Nowcast models are generally reduced-form representations of the economy. They are not structural models.

- As such, they are ideal for producing more timely forecasts of a low frequency variable, but they *cannot explain how a forecast may change in response to a policy change*.

- For this, one needs to use a structural VAR/DSGE model.
  - But allowing for mixed frequency variables in a structural model is theoretically possible, but quite challenging (e.g., given the need for identification restrictions for the SVAR, or calibration and dimensionality issues in DSGE).

Approaches used to prepare a nowcast

- Bridge equations (e.g., Ferrara & Simoni, 2010)
- MIDAS (Mixed Data Sampling) (e.g., Clements & Galvao, 2007)
- Mixed Frequency VARs (e.g., Mariano & Murasawa (2010), Giannone, Reichlin & Small (2008))
- Mixed Frequency Factor Models (e.g., Giannini, Reichlin & Small, 2008)
- Factor-augmented autoregressive models (e.g., Grui & Lysenko, 2017)
- Machine learning (e.g., Barhoumi et al (2020), Woloszko (2020))
- Accounting-based tracking models (e.g., Higgins, 2014)

Approaches used to prepare a nowcast

- One can classify nowcasting models into univariate (e.g., Bridge, MIDAS) or multivariate (e.g., VAR, Dynamic Factor models)
- Multivariate approaches also model the feedback effect of the low frequency variable onto the high frequency indicators.
- For multivariate models, one can use either a regression-based approach or a state-space representation combined with the Kalman filter.
  - the latter is optimal, in the sense of producing minimum MSE nowcasts given a correct model specification
  - it is computationally demanding, when the frequency mismatch is large, a, for example, with daily/quarterly data.

  It is often more convenient to start with the regression-based approach, which is also adopted in univariate models.

Econometric Problem

- Depending on the frequency mismatch (target frequency and the RHS frequency), the number of parameters to be estimated in a nowcasting exercise can grow rather large very quickly.
- The nowcasting regression may have too many parameters to estimate precisely, leading to wide confidence bands.
- For example, suppose we want to use daily interest rates to update the forecast of quarterly GDP. Each day in the quarter could be used as a 'daily' quarterly series, resulting in additional 91 regressors.

Econometric Problem

- Basic problem: the large number of parameters that may need to be estimated.
- One solution: to convert the high frequency series (e.g., daily) to the low frequency (e.g., quarterly) using the sum (or average) of the HF variables.
  - For example, 91 daily series could be reduced to 1 by taking sums or averaging, drastically reducing the number of parameters to estimate, but at the expense of forcing every day of the quarter to have the same effect on the quarterly series. This is the BRIDGE approach.
- Another approach: to propose that the individual parameters to be estimated for the HF variables follow a function that depends on *far fewer* parameters. This is the MIDAS approach.
- Alternatively: One could summarize the many regressors by taking their principal components (e.g., Australian paper) or modelling them together with the target variable using a dynamic factor model/Kalman Filter (e.g., NY FED model).

Country Examples

1. St. Louis Federal Reserve Study
2. Queensland Treasury (MIDAS)
3. National Bank of Belgium study (Dynamic Factor Model)
4. Atlanta Fed Nowcast
5. BOE Nowcast

Forecasting with Mixed Frequencies: A paper from Federal Reserve Bank of St-Louis

Armesto *et. al.* (2010) Goal: Survey several common methods for dealing with mixed-frequency data and analyze their forecast performance

Method: Simple time averaging, a step-weighting function, and exponential Almon polynomial MIDAS

Target Nowcasts:

(i) *quarterly GDP growth*, using its own lags and monthly employment growth rates
(ii) *monthly inflation*, using its own lags and daily interest rates
(iii) *monthly industrial production (IP) growth*, using its own lags and daily interest rates
(iv) *monthly employment growth*, using its own lags and daily interest rates

Forecasting with Mixed Frequencies: A paper from Federal Reserve Bank of St-Louis

Which method is best for forecasting purposes? As one might expect, the results are data dependent.

- The performances of different time-aggregation approaches vary—there does not appear to be a universal rule.
- For some variables, simply averaging the higher-frequency data produces no discernible disadvantage.
- In other cases, explicitly modeling the flow of data (e.g., MIDAS) may be more beneficial, especially for constructing intra-period forecasts.

MIDAS regressions and daily nowcast of quarterly trend Australian real GDP growth

Anthonisz, M. (2018) Goal: Nowcast real-time(daily) GDP with over 60 HF variables, which are too many for MIDAS to handle.

Rather than discard variables, principal component analysis (PCA) is first used to extract the common trends among the dependent variables.

Note: Nowcasts of GDP will be quarterly, like the frequency of the official GDP release. Nowcasting do not produce a forecast of daily GDP.

NOTE: Rather than PCA, 3-Pass Regression Filters (3PRF) could be used to extract the factors (more on this in Lecture 4).

MIDAS regressions and daily nowcast of quarterly trend Australian real GDP growth

Common factors are calculated at the lowest frequency of the underlying data (in this case, quarterly). They are complemented in the model by daily financial data which may have some explanatory power to nowcast the quarterly change in Australian GDP.

Daily data include the trade weighted exchange rate; the three-year sovereign bond yield.

MIDAS regressions and daily nowcast of quarterly trend Australian real GDP growth

Example of the rolling process to update nowcasts

Assume it is currently 1 July 2018.

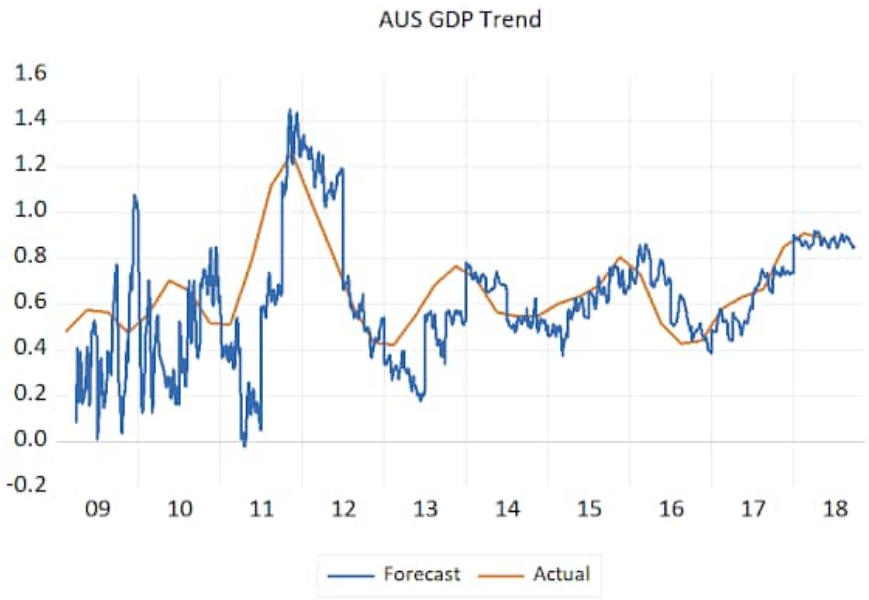
- We want to forecast Q3 2018 GDP using one period lags of GDP and the common factors estimated earlier via principal component analysis (PCA)
    - Common factors are quarterly representations of conditions with respect to labor markets and capital investment as well as measures of current and future economic activity.
    - Daily data include bond yields and the trade-weighted exchange rate.
- The model is estimated using historical data from Q2 1993 until Q2 2018 (on 1 July, we have data to 30 June).
    - Specify the equation of each quarter's GDP growth as a function of the previous quarter's outcomes for the quarterly variables and of the last 45 days' worth of values for bond yields and the exchange rate ending on the last day of the previous quarter.

MIDAS regressions and daily nowcast of quarterly trend Australian real GDP growth

- Once the model is estimated, use the 45 daily values for bond yields and the exchange rate from May to June 2018 to forecast Q3 GDP.
- On 2 July 2018, we have one more observation for the daily series. Update the forecast of GDP by estimating a new model on historical data that used 44 days from the previous quarter and the first day from the current quarter, and then forecast Q3 GDP.
- On 3 July 2018, update the forecast by estimating on 43 days of the previous quarter and the first 2 days from the current quarter. And so on.

→ This gives a forecast of quarterly GDP that is updated daily.
→ This does not make it a forecast of daily GDP as it is a quarterly variable.
→ It is a method to forecast GDP using current (now) data and update this forecast daily.

MIDAS regressions and daily nowcast of quarterly trend Australian real GDP growth



A mixed–frequency dynamic factor model for nowcasting consumer confidence

Policymakers, firms, and investors closely monitor traditional survey–based consumer confidence indicators as an important piece of economic information.

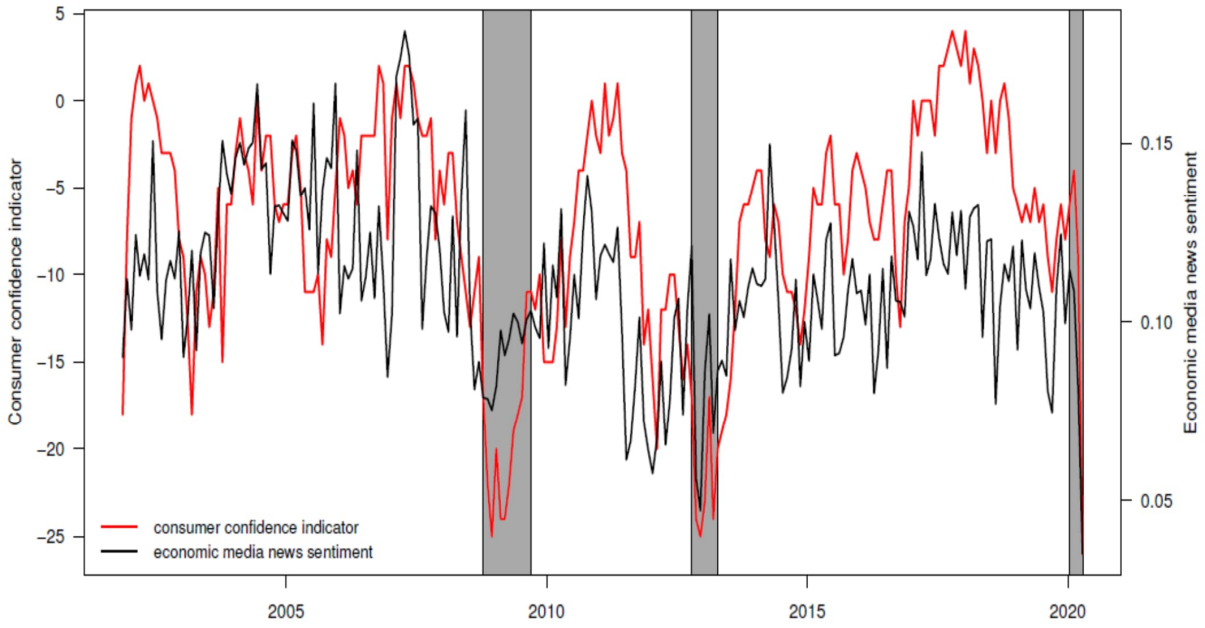
Algada et. al. (2021) propose a framework--a mixed-frequency Dynamic Factor Model--to nowcast the monthly survey-based consumer confidence indicator with the daily sentiment embedded in economic media news articles. See National Bank of Belgium.

Results:

- Economically significant accuracy gains are obtained by including daily news sentiment.
- The daily average economic media news sentiment is useful for constructing a latent coincident consumer confidence index.
- The recent COVID-19 pandemic serves as an illustration of the usefulness of the mixed-frequency model in times of rapid changes.

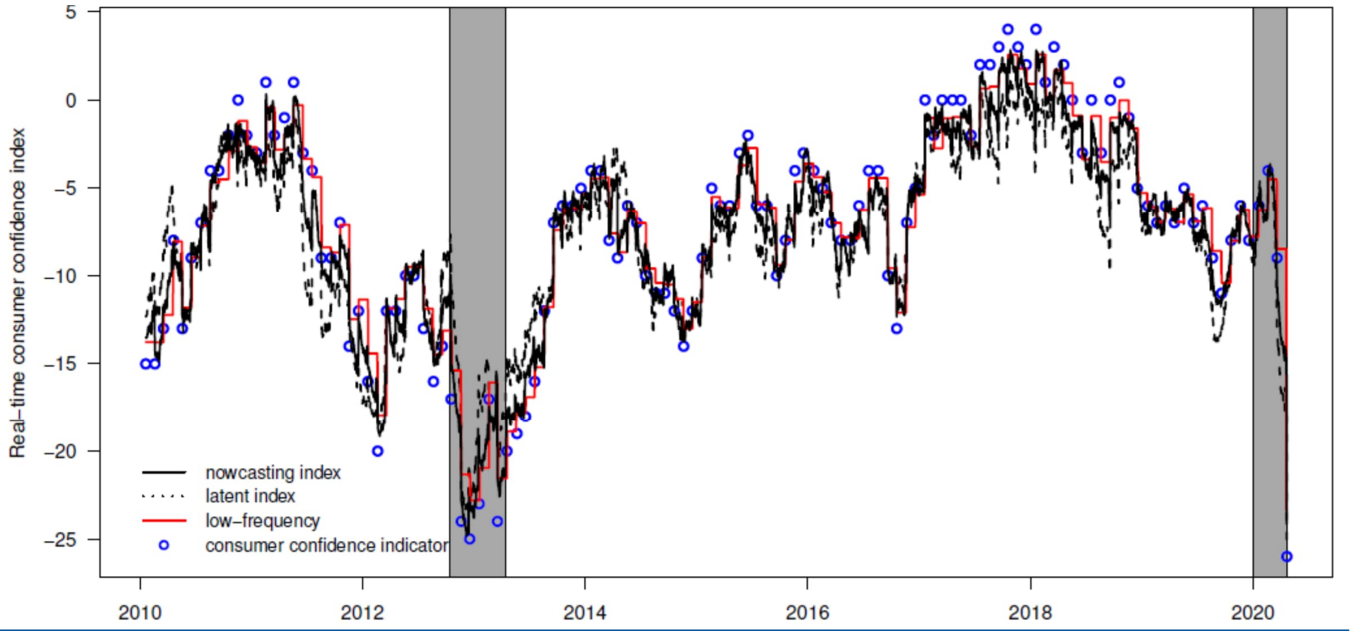
A mixed–frequency dynamic factor model for nowcasting consumer confidence

Monthly economic media news sentiment and survey-based consumer confidence indicator over the period November 2001 until April 2020.



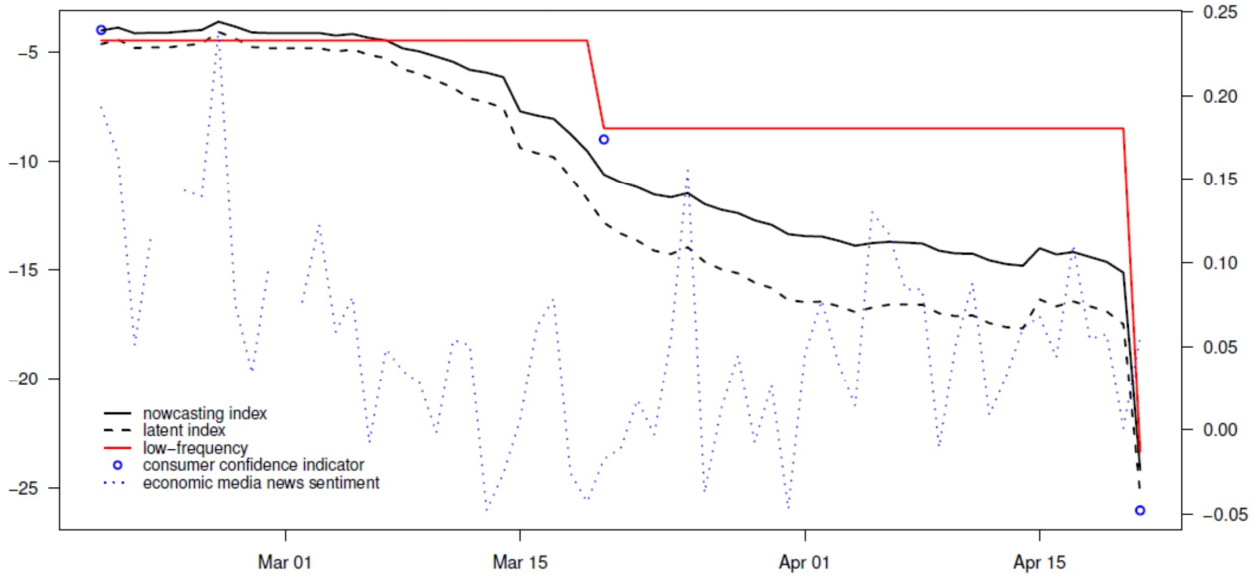
A mixed–frequency dynamic factor model for nowcasting consumer confidence

Daily real time estimates of the mixed-frequency models, one-step ahead forecasts of the low-frequency (LF) model, and the monthly survey-based consumer confidence indicator as measured by the National Bank of Belgium over January 2010 - April 2020.



A mixed–frequency dynamic factor model for nowcasting consumer confidence

Daily one-step ahead forecasts of the LF model, real time estimates of the mixed-frequency models, the survey-based consumer confidence indicator, and economic media news sentiment over 19 February 2020 - 21 April 2020.



Atlanta Fed GDPNow

Definition & Purpose

GDPNow is a “nowcasting” tool developed by the Federal Reserve Bank of Atlanta that provides real-time, model-driven estimates of current-quarter U.S. real GDP growth – well before the official Bureau of Economic Analysis (BEA) release.

Modeling Methodology

It aggregates 13 GDP subcomponents (e.g., personal consumption, investment, trade, government spending) using a mix of Bridge Equations, Factor Models, and Bayesian Vector Autoregressions. This approach aligns well with the chain-weighting methods used by the BEA, ensuring structural consistency with official GDP accounting.

Update Frequency & Data Inputs

GDPNow updates approximately 6-7 times per month, incorporating data from sources like the Census Bureau, Bureau of Labor Statistics, and other agencies.

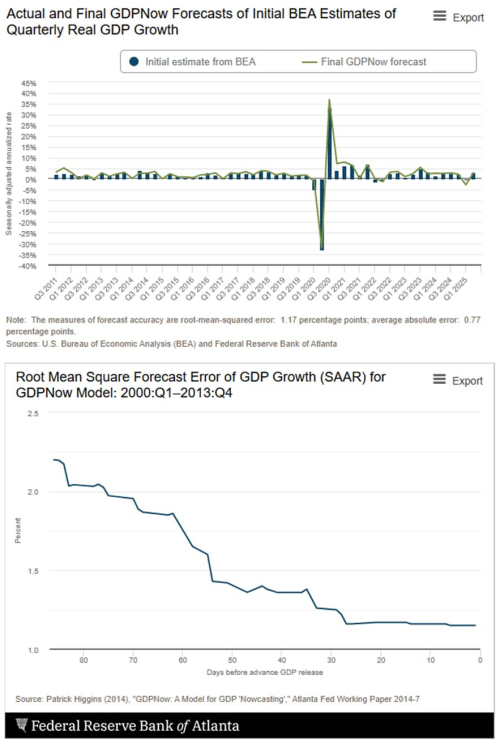
Atlanta Fed GDPNow

Why Trust this Model?

- Transparent: full methodology and source code are publicly available
- Consistent: uses the same expenditure framework as the BEA
- Data-Driven: free of subjective adjustments or political influence
- Track Record: historically competitive with private-sector forecasts, especially as more data accumulate during the quarter

Limitations & Accuracy Variability

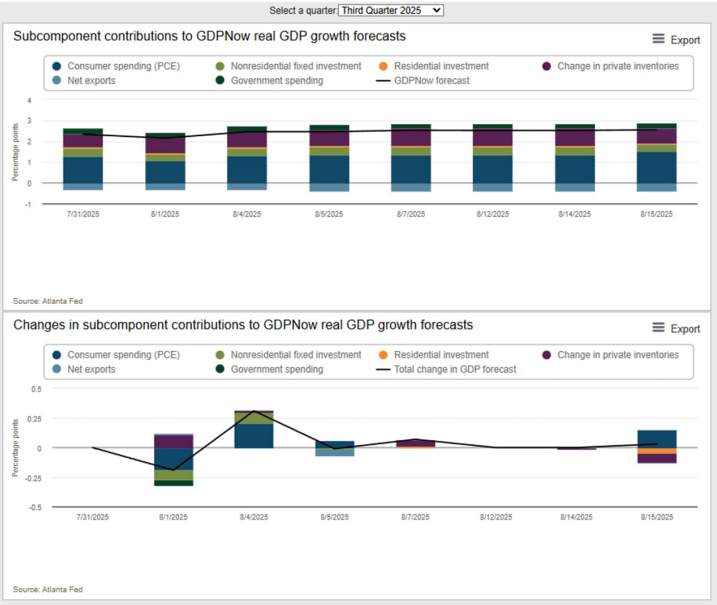
During the COVID-19 pandemic, GDPNow’s performance declined, with the model notably struggling to capture rapid shifts in services consumption and other subcomponents.



Atlanta Fed GDPNow

How to Read the Output?

- Main output: an annualized real GDP growth rate (%) for the current quarter.
- Subcomponent Contributions Charts: shows the evolution of the estimate over time as new data are added, often alongside consensus forecasts for comparison



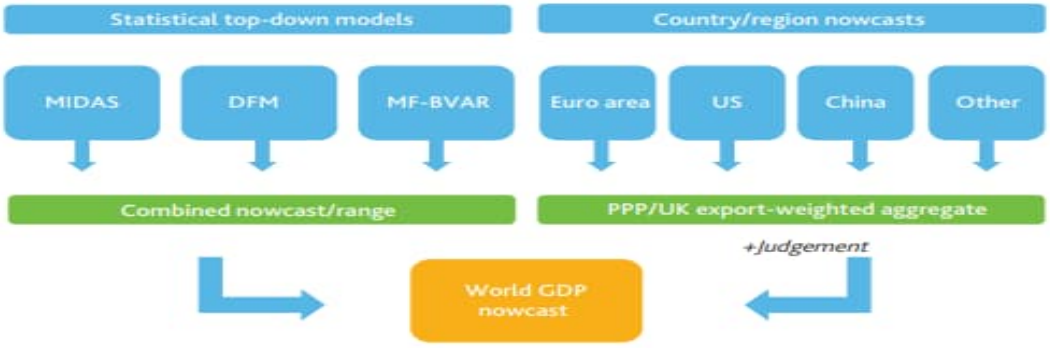
Source: Federal Reserve Bank of Atlanta. "GDPNow." *Center for Quantitative Economic Research*, updated regularly. Available at: https://www.atlantafed.org/cqer/research/gdpnow

Bank of England and Nowcasts

Bank of England (BoE) nowcasts global GDP growth using a suite of models and wide range of high-frequency data. Nowcasts are a key output into the Monetary Policy Committee's assessment of the UK economic outlook.

The models in the nowcasting suite:

Figure 1 Country-level and global top-down nowcasts inform the Bank's near-term outlook for world activity



Source: Kindberg-Hanlon and Sokol (2018)

Bank of England and Nowcasts

BoE's nowcast models use various types of data:

Soft data: Surveys of current economic conditions, (e.g., purchasing managers' indices (PMIs))

Hard data: data on trade, retail sales and industrial production (IP). These indicators are produced on a monthly basis by national statistical authorities and ultimately feed into the production of GDP statistics.

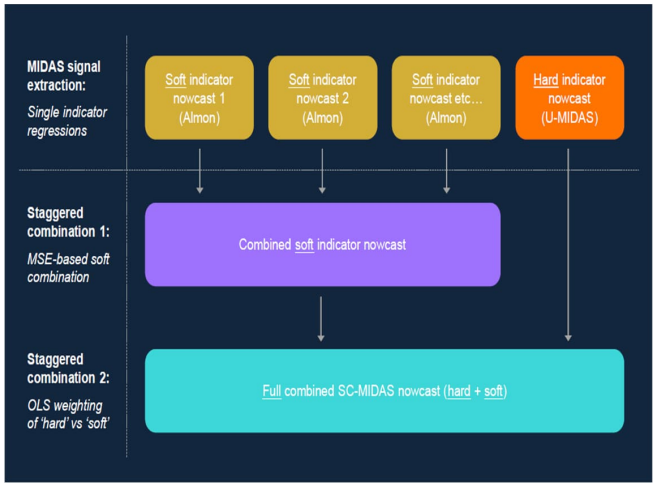
Market data: equities, interest rates, exchange rates and commodity prices provide important information on confidence and economic sentiment.

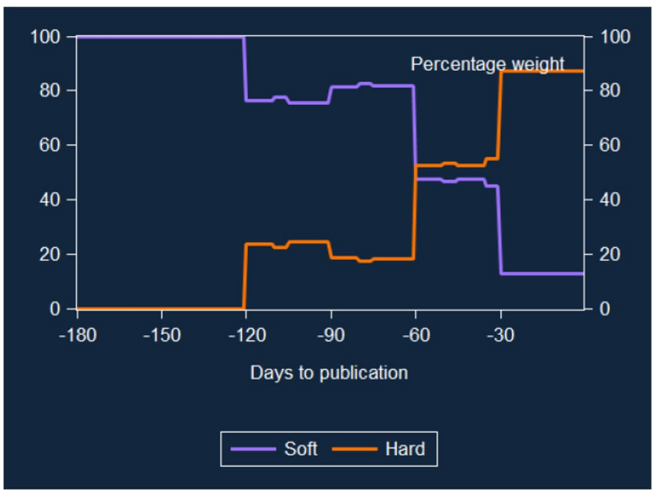
- One of the advantages is timeliness; a downside is that not all factors that affect asset and commodity prices are related to activity levels, making them noisier than other types of data.

Bank of England and Nowcasts

The Staggered-Combination MIDAS (SC-MIDAS) approach combines both soft and hard data.

Weights shift flexibly. Early in cycle, 100% soft data. Higher weight towards hard data by the end.

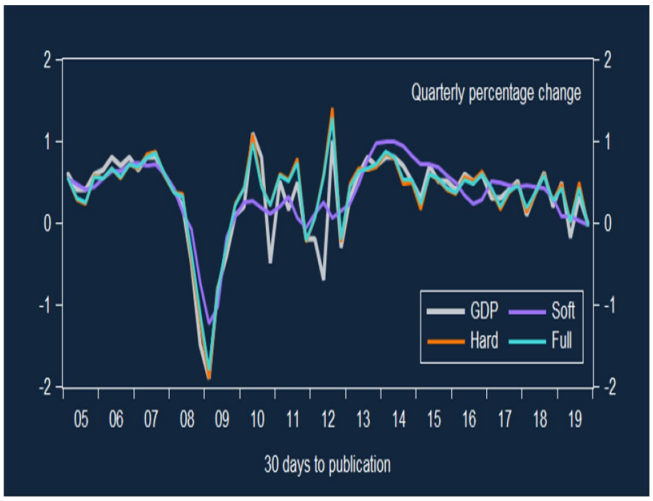




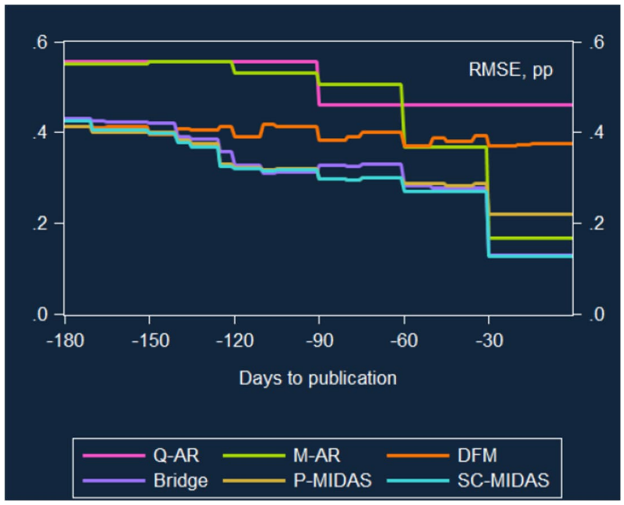
Source: Moreira (2025)

Bank of England and Nowcasts

Out-of-sample nowcast snapshot: full nowcast (soft+hard) closely tracks growth.



Horserace results favor SC-MIDAS



Source: Moreira (2025)

Uncertainty around Nowcasts

- Uncertainty is inherent in nowcasting: no approach can predict GDP growth perfectly ahead of its publication. Nowcast errors are to be expected, resulting in a range of possible outcomes.
- Past errors can offer a guide to the distribution of possible outcomes surrounding any given nowcast — but past errors may not be fully representative of the true underlying uncertainty, particularly with limited samples.
- Nowcast uncertainty varies across models and through the data cycle. The root mean squared errors provide a summary measure of that uncertainty: a higher RMSE normally indicates a wider dispersion of possible outcomes (wider prediction interval around the central nowcasts).

Conclusion

- A nowcast is a forecast of the present or near future. Nowcasting can incorporate newly-released data into the forecast in real time.
- The challenges arise from unbalanced data (mixed frequency, ragged-edge problems).
  - Frequency mismatch can quickly lead to many parameters to be estimated.
- Nowcast models can deal with those challenges. There is no rule on the best forecasting models.
- Need to remain alert to model shortcomings:
  - Adjust the model when needed; models should be under constant development.
  - Consider shifting to alternative models promptly.

Conclusion

- Forecasting are never mechanical constructs from a particular model; need to incorporate best judgment about developments in the economy and the monetary transmission mechanism.
- Do not rely on a single model: suites of models and judgment should be used to inform "official" forecasts.
  The mapping from indicators to official GDP estimates is imperfect so it is important to pay attention to the confidence bands surrounding any given estimate.
- *As a caveat, nowcasting models are typically non-structural models → nowcasting is not well suited to analyze the impact of policy changes.*

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